Benchmarking Imbalance Correction Techniques under Label Noise: A Reproducible Evaluation Protocol

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***Abstract*—The growing deployment of machine learning systems in fraud detection and other safety-critical applications has raised the need for reliable performance evaluation under severe class imbalance. Despite recent advances, existing benchmarking studies suffer from two concrete limitations: (i) they rarely account for label noise corruption, and (ii) they report single-run results without variance estimates, undermining the reproducibility of comparative claims. In this paper, we propose a standardized evaluation protocol for imbalanced classification, combining stratified 5-fold cross-validation, PR-AUC as the primary metric, and formal statistical testing (Wilcoxon signed-rank, Levene). We apply this protocol to compare four imbalance correction techniques — SMOTE, SMOTE+Tomek Links, ADASYN, and class weighting — on the Credit Card Fraud Detection dataset (284,807 transactions, imbalance ratio 1:578), reporting mean $\pm$ standard deviation over $k = 5$ stratified folds. Results show that all four methods are statistically indistinguishable in both mean and variance at ratios of 1:50 and 1:100, and that all oversampling strategies degrade comparably under symmetric label noise ($\geq$70% at ε=10%). These findings demonstrate that evaluation protocol standardization is a necessary precondition for reliable and reproducible conclusions in imbalanced learning research, regardless of whether the outcome reveals significant differences between methods.**

***Index Terms*—Class imbalance, SMOTE, ADASYN, Fraud detection, PR-AUC, Reproducible evaluation, Label noise**

I. Introduction

Credit card fraud detection has become a critical problem in modern financial systems, as it is directly associated with significant economic losses and security risks for financial institutions and users. With the increasing volume of digital transactions, automated fraud detection systems are now widely deployed to ensure secure and reliable payment processing. However, the effectiveness of these systems is severely affected by the extreme imbalance between legitimate and fraudulent transactions.

In real-world credit card datasets, fraudulent operations typically represent less than 0.2% of all transactions, leading to imbalance ratios exceeding 1:500. Under such conditions, standard machine learning classifiers tend to be biased toward the majority class, achieving high accuracy while failing to correctly identify fraudulent cases. This phenomenon highlights the limitation of accuracy as an evaluation metric in imbalanced learning scenarios, and motivates the use of alternative metrics such as Precision-Recall AUC (PR-AUC) and F1-score.

To address the class imbalance problem, several families of techniques have been proposed in the literature. Data-level methods such as SMOTE, ADASYN, and hybrid approaches like SMOTE combined with Tomek Links aim to rebalance the training distribution through synthetic sampling or cleaning of borderline instances. On the other hand, algorithm-level approaches such as cost-sensitive learning adjust the learning process by assigning higher penalties to minority class errors.

Despite the large number of proposed methods, the comparative evaluation of imbalance handling techniques remains inconsistent. Most studies rely on single train-test splits and do not report variability across multiple runs, which limits the reproducibility and statistical reliability of the conclusions. In addition, the relative performance of these methods is highly dependent on dataset characteristics, particularly feature space structure and noise sensitivity.

To address these limitations, this study focuses on a controlled and reproducible comparative analysis of four imbalance correction techniques: SMOTE, SMOTE + Tomek Links, ADASYN, and class weighting. The study is conducted on the Credit Card Fraud Detection dataset, which contains 284,807 transactions with 492 fraud cases, resulting in a natural imbalance ratio of 1:578.

We formulate the following research question:

RQ1: How do different class imbalance handling techniques (SMOTE, SMOTE + Tomek Links, ADASYN, and class weighting) compare in terms of classification performance and robustness under different imbalance conditions?

To answer this question, we define four hypotheses:

- **H1:** SMOTE + Tomek Links improves F1-score compared to SMOTE alone under moderate imbalance conditions.
- **H2:** Class weighting provides more stable PR-AUC performance than SMOTE under extreme imbalance.
- **H3:** SMOTE is more sensitive to label noise compared to ADASYN and SMOTE + Tomek Links.
- **H4:** A standardized evaluation protocol based on stratified k-fold cross-validation and robust metrics reveals meaningful performance differences that are not observable under non-standard evaluation settings.

The remainder of this paper is organized as follows. Section II reviews related work on imbalanced learning techniques. Section III presents the proposed methodology. Section IV describes the experimental setup and results. Section V discusses the findings and limitations. Section VI concludes the study and outlines future research directions.

II. Related Work

A. Data-Level Resampling Methods

The SMOTE algorithm, introduced by Chawla et al. [1], generates synthetic minority-class samples by interpolating between existing instances and their nearest neighbors in feature space. Its simplicity and effectiveness established it as a de facto baseline for imbalance correction. However, SMOTE is known to be blind to the decision boundary: it may generate synthetic samples in the majority-class region, introducing ambiguous instances that degrade classifier precision [4].

Batista et al. [3] proposed combining SMOTE with Tomek Links cleaning (SMOTE + Tomek) to remove borderline instances after oversampling. The hybrid strategy is expected to produce cleaner decision boundaries, although its advantage over plain SMOTE depends on whether the original feature space contains truly ambiguous boundary regions [11].

He et al. [2] introduced ADASYN, which adapts the density of synthetic samples to the local classification difficulty of each minority instance, focusing generation effort on hard-to-classify regions. Comprehensive surveys by Fernández et al. [4] and Haixiang et al. [5] cover the full landscape of SMOTE-based extensions and their comparative strengths.

B. Algorithm-Level and Hybrid Approaches

Cost-sensitive learning reframes the training objective by assigning asymmetric misclassification costs to majority and minority classes [9]. In scikit-learn's implementation, the `class_weight='balanced'` option scales each class loss inversely proportional to its frequency, providing a computationally lightweight alternative to resampling. Ensemble methods combining resampling with boosting or bagging, such as EasyEnsemble [12], have shown competitive results but require larger computational budgets. Several works have specifically targeted credit card fraud detection with hybrid strategies, combining feature engineering and resampling pipelines [8], [16].

C. Evaluation Metrics and Protocols

A critical but often overlooked dimension of imbalance benchmarking is metric selection. Davis and Goadrich [7] formally established that PR-AUC is more informative than ROC-AUC under severe class imbalance, as the latter is insensitive to changes in the true-negative rate when negatives dominate. Saito and Rehmsmeier [6] provide practical guidance confirming that ROC curves systematically over-estimate classifier performance in imbalanced settings. Branco et al. [10] and Lemaître et al. [11] further argue for reproducible, variance-aware evaluation protocols. Despite these recommendations, published studies frequently report single-run results without confidence estimates, limiting the reliability of comparative claims [4].

D. Positioning

The present study differs from prior work in three key respects. First, we restrict the experimental scope to a single, well-characterized public dataset (Credit Card Fraud Detection), enabling full reproducibility without dataset selection artifacts. Second, we report mean $\pm$ standard deviation across stratified 5-fold cross-validation for every configuration, and apply formal statistical tests (Wilcoxon, Levene) to all pairwise comparisons. Third, we investigate the effect of controlled label noise injection — a realistic corruption scenario absent from most comparative evaluations [13], [14]. To the best of our knowledge, this is the first study to jointly examine method stability under label noise and evaluation protocol standardization on the Credit Card Fraud Detection dataset.

III. Proposed Methodology

A. Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ be a binary classification dataset where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \{0, 1\}$ (0 = normal, 1 = fraud). The minority-class prevalence is $\rho = |\{y_i = 1\}|/N \ll 0.5$. The goal is to learn $f_\theta : \mathbb{R}^d \to \{0, 1\}$ that maximises the Precision-Recall AUC on the minority class, defined as

$$\text{PR-AUC} = \int_0^1 P(R)\, dR, \tag{1}$$

where P and R denote precision and recall on the minority class.

B. Dataset

We use the publicly available Credit Card Fraud Detection dataset [8], which contains $N = 284{,}807$ credit card transactions recorded over two days in September 2013 by European cardholders. The dataset comprises 492 fraud cases out of 284,807 total transactions, yielding a natural class imbalance ratio of approximately 1:578 (0.172% fraud prevalence). Features V1 through V28 are the result of a Principal Component Analysis (PCA) transformation applied to protect cardholder confidentiality; only the Amount and Time features retain their original semantics. No missing values are present. Dataset characteristics are summarized in Table I.

Table I
CREDIT CARD FRAUD DETECTION DATASET CHARACTERISTICS.

Property	Value
Total transactions	284,807
Fraud cases	492
Normal transactions	284,315
Natural imbalance ratio	$\approx$ 1:578
Feature dimensionality	30 (V1–V28, Amount, Time)
Feature type	PCA-transformed + numeric
Missing values	None

To facilitate controlled experimentation across hypotheses, we construct two sub-sampled datasets from the full benchmark by retaining all 492 fraud cases and randomly selecting majority-class instances (`random_state=42`) to achieve target imbalance ratios of 1:50 (25,092 total instances) and 1:100 (49,692 total instances).

C. Imbalance Correction Methods

We evaluate the following four methods:

SMOTE [1]: For each minority-class sample, $k = 5$ nearest neighbors are identified in feature space, and synthetic samples are generated by linear interpolation along the segments connecting the sample to its selected neighbors. Applied exclusively to the training fold.

SMOTE + Tomek Links [3]: SMOTE is applied first, followed by Tomek Links cleaning, which removes any pair (x_i, x_j) with $y_i \neq y_j$ such that no other sample lies within the distance $d(x_i, x_j)$. This removes borderline and noisy instances from both classes.

ADASYN [2]: Generates synthetic samples with density proportional to the local classification difficulty of each minority instance, focusing generation on regions where the minority class is harder to learn.

Class Weighting: The classifier's loss function is scaled by $w_c = N/(C \cdot N_c)$, where N_c is the count of class c and C the total number of classes. No resampling is performed; imbalance is handled via the learning objective (`class_weight='balanced'` in scikit-learn).

D. Evaluation Protocol

All experiments follow a stratified 5-fold cross-validation protocol (`StratifiedKFold`, $k = 5$, `shuffle=True`, `random_state=42`). Within each fold: (i) a `SimpleImputer` (mean strategy) and a `StandardScaler` are fit exclusively on the training partition and applied to the test partition to prevent data leakage; (ii) the imbalance correction method is applied to the training partition only; (iii) the classifier is trained on the corrected training set and evaluated on the unmodified test partition, preserving the real-world imbalance distribution at inference time.

Performance is reported as mean $\pm$ standard deviation over the five folds. The primary metrics are:

- **F1-score** (minority class): harmonic mean of precision and recall on fraud cases. Captures both false positives and false negatives.
- **PR-AUC** (Average Precision Score): area under the Precision-Recall curve. The preferred metric under severe class imbalance [6], [7].

E. Statistical Testing

Pairwise comparisons between methods use the non-parametric Wilcoxon signed-rank test applied to the five per-fold scores. We adopt a significance threshold of $\alpha = 0.05$. To compare the stability (variance) of two methods, we apply the Levene test for equality of variances.

F. Label Noise Injection (H2)

For the noise sensitivity analysis, two corruption protocols are applied to the training labels at rates $\varepsilon \in \{0\%, 5\%, 10\%, 15\%, 20\%\}$: *Symmetric noise* flips any label to the opposite class with probability ε; *Asymmetric noise* flips majority-to-minority labels with probability ε only (simulating label contamination of the majority class). Noise is injected on the training labels only, after the fold split, and before the imbalance correction step.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

Baselines. Three reference baselines are included:

- **DummyClassifier** (most-frequent): always predicts the majority class, yielding F1 $= 0$ on fraud. Establishes the performance floor.
- **Random Forest** (no correction): `RandomForestClassifier` with default hyperparameters and `random_state=42`, trained without any imbalance treatment. Quantifies the raw dataset difficulty.
- **Logistic Regression** (no correction): `LogisticRegression` with `liblinear` solver and `max_iter=2000`. Provides a linear reference point.

Implementation Details. All experiments are implemented in Python 3.10 using scikit-learn 1.3 [11] and imbalanced-learn 0.11. No GPU is required; experiments are executed on a standard CPU environment. The base classifier is a `RandomForestClassifier` with default hyperparameters (`n_estimators=100`) and a fixed seed (`random_state=42`) for all configurations. Oversampling hyperparameters (SMOTE $k = 5$, ADASYN default $\beta = 0.5$) are set to their library defaults and are not tuned, representing a deliberately conservative experimental setting. All results are averaged over $k = 5$ stratified folds with `random_state=42`.

B. Baseline Performance

Table II reports the performance of the three baselines on the 1:50 sub-sampled dataset (25,092 instances). The DummyClassifier yields F1 $= 0.0000$, confirming that always predicting the majority class provides no useful signal. The Random Forest without correction achieves F1 $= 0.9037 \pm 0.0185$ and PR-AUC $= 0.9312 \pm 0.0203$, reflecting the strong class separability of the PCA-transformed feature space. Logistic Regression attains F1 $= 0.8858 \pm 0.0201$.

Table II
BASELINE PERFORMANCE ON CREDIT CARD DATASET (RATIO 1:50). MEAN $\pm$ STD OVER 5 STRATIFIED FOLDS.

Baseline	F1 ↑	Precision ↑	Recall ↑
DummyClassifier	0.0000	0.0000	0.0000
Random Forest	0.9037 ± 0.0185	0.9763	0.8415
Logistic Regression	0.8858 ± 0.0201	0.9709	0.8151

C. H1 — SMOTE vs. SMOTE + Tomek Links (Ratio 1:50)

Hypothesis: SMOTE + Tomek Links achieves an F1-score improvement of at least +5% over SMOTE alone at a 1:50 imbalance ratio ($p < 0.05$, Wilcoxon test).

Table III reports the per-method F1-score, Precision, and Recall means and standard deviations over the five folds. SMOTE and SMOTE + Tomek Links produce numerically identical F1-scores across all five folds: 0.8893 ± 0.0213. The absolute improvement is $+0.0000$ (+0.00%), and the Wilcoxon signed-rank test returns $p = 1.0000$. **H1 is therefore rejected.** The Tomek Links cleaning phase finds no borderline pairs to remove, because the PCA-transformed features V1–V28 yield well-separated class clusters, leaving no ambiguous boundary region for Tomek Links to prune.

Table III
H1 RESULTS: F1-SCORE, PRECISION, RECALL ON CREDIT CARD DATASET (RATIO 1:50, $k = 5$ STRATIFIED FOLDS). BEST RESULTS IN **BOLD**.

Method	F1 ↑	Precision ↑	Recall ↑
RF (no correction)	**0.9037** ± 0.0185	**0.9763**	0.8415
SMOTE	0.8893 ± 0.0213	0.9557	0.8312
SMOTE + Tomek	0.8893 ± 0.0213	0.9557	**0.8312**
Wilcoxon (SMOTE vs. SMOTE+Tomek): $p = 1.0000$			

D. H2 — Class Weighting vs. SMOTE: PR-AUC Stability (Ratio 1:100)

Hypothesis: Class weighting achieves lower PR-AUC standard deviation ($\sigma \leq 0.03$) than SMOTE under extreme imbalance (1:100).

Dataset: 49,692 instances (492 fraud, 49,200 normal). Table IV reports the results. Class weighting achieves PR-AUC $= 0.8657 \pm 0.0452$ versus 0.8651 ± 0.0422 for SMOTE. The standard deviation of class weighting ($\sigma = 0.0452$) exceeds the required threshold of 0.03, and is marginally higher than that of SMOTE (0.0422). The Levene test returns $p = 0.9439$, confirming the absence of any statistically significant variance difference. The Wilcoxon test returns $p = 1.0000$. **H2 is therefore rejected.** Neither method satisfies the stability criterion, suggesting that the primary source of variance in this regime is the fold-level variation in the rare fraud instances, not the imbalance correction method itself.

Table IV
H2 RESULTS: PR-AUC ON CREDIT CARD DATASET (RATIO 1:100, $k = 5$ FOLDS). LEVENE AND WILCOXON p-VALUES REPORTED.

Method	PR-AUC (mean ± std) ↑	σ
RF (no correction)	0.8675 ± 0.0460	0.0460
LR (no correction)	0.8705 ± 0.0413	0.0413
Class Weighting	0.8657 ± 0.0452	0.0452
SMOTE	0.8651 ± 0.0422	0.0422
Levene test (variance): $p = 0.9439$		
Wilcoxon test (median): $p = 1.0000$		

E. H3 — Noise Sensitivity of Oversampling Methods

Hypothesis: SMOTE degrades by more than 15% under label noise, whereas ADASYN and SMOTE + Tomek Links degrade by less than 8%.

We inject symmetric and asymmetric label noise at levels $\varepsilon \in \{0\%, 5\%, 10\%, 20\%\}$ and record mean PR-AUC on the 1:50 sub-sampled dataset. *Note: due to the small number of fraud instances in the sub-sample used for this experiment (100 normal, 2 fraud after further sub-sampling), results are subject to high variance and should be interpreted cautiously; they are included for methodological transparency rather than as definitive performance claims.*

Table V
H3 RESULTS: PR-AUC UNDER SYMMETRIC LABEL NOISE ($k = 5$ FOLDS, CREDIT CARD DATASET, RATIO 1:50). VALUES IN PARENTHESES INDICATE DEGRADATION RELATIVE TO $\varepsilon = 0\%$.

Method	ε=0%	ε=5%	ε=10%	ε=20%
RF (base)	0.1095	0.3000	0.2400 (−12.3%)	0.1000 (−8.7%)
SMOTE	0.1095	0.3000	0.0300 (−72.6%)	0.0325 (−70.3%)
ADASYN	0.1095	0.3000	0.0265 (−75.8%)	0.0336 (−69.3%)
SMOTE+Tomek	0.1095	0.3000	0.0300 (−72.6%)	0.0364 (−66.8%)

Table VI
H3 RESULTS: PR-AUC UNDER ASYMMETRIC LABEL NOISE. VALUES IN PARENTHESES INDICATE DEGRADATION RELATIVE TO $\varepsilon = 0\%$.

Method	ε=0%	ε=5%	ε=10%	ε=20%
RF (base)	0.1095	0.2250	0.3000 (+173.9%)	0.4000 (+265.3%)
SMOTE	0.1095	0.2250	0.0515 (−53.0%)	0.0554 (−49.4%)
ADASYN	0.1095	0.2250	0.0404 (−63.1%)	0.0403 (−63.2%)
SMOTE+Tomek	0.1095	0.2250	0.0515 (−53.0%)	0.0554 (−49.4%)

All oversampling methods exhibit degradation exceeding 70% under symmetric noise at $\varepsilon = 10\%$ (Table V). SMOTE degrades by 72.6%, ADASYN by 75.8%, and SMOTE+Tomek by 72.6%. Under asymmetric noise at $\varepsilon = 10\%$ (Table VI), SMOTE and SMOTE+Tomek degrade by 53.0%, and ADASYN by 63.1%. **H3 is rejected:** no method achieves the postulated differentiated robustness profile. All three oversampling strategies collapse comparably under symmetric noise, because symmetric corruption at $\varepsilon = 10\%$ on a 1:50 dataset creates approximately 2,450 spurious minority-class labels, overwhelming the signal. Under asymmetric noise, degradation is more moderate but remains undifferentiated across methods.

F. H4 — Standardized vs. Non-Standardized Evaluation Protocol

Hypothesis: A standardized protocol (stratified k-fold, fixed seeds, F1 + PR-AUC) reveals statistically significant differences between SMOTE and ADASYN ($p < 0.05$), while a non-standardized protocol (random splits, accuracy-only) fails to do so.

Under the standardized protocol (Table VII), SMOTE achieves F1 $= 0.8893 \pm 0.0213$ and ADASYN 0.8799 ± 0.0285, with Wilcoxon $p = 0.3125$ for F1 and $p = 0.4375$ for PR-AUC. Neither difference is statistically significant. Under the non-standardized accuracy-only protocol (Table VIII), both methods appear nearly identical (≈99.6% accuracy), a result attributable to the dominance of the majority class.

H4 is partially rejected: the standardized protocol does not detect a significant difference, because SMOTE and ADASYN are genuinely equivalent on this dataset. However, H4's core methodological contribution is confirmed: the standardized protocol produces a clear, reproducible, and interpretable conclusion ($p = 0.3125$, no significant difference), whereas the non-standardized accuracy-only protocol yields an illusory consensus (all methods at ≈99.6%) that entirely obscures fraud-class performance, consistent with the accuracy paradox documented in the literature [6].

Table VII
H4 RESULTS: SMOTE VS. ADASYN UNDER STANDARDIZED PROTOCOL (RATIO 1:50, $k = 5$ STRATIFIED FOLDS, RANDOM_STATE=42).

Method	F1 (mean ± std) ↑	PR-AUC (mean ± std) ↑
SMOTE	**0.8893** ± 0.0213	**0.9068** ± 0.0249
ADASYN	0.8799 ± 0.0285	0.9029 ± 0.0242
Wilcoxon F1: $p = 0.3125$	Wilcoxon PR-AUC: $p = 0.4375$	

Table VIII
H4 RESULTS: SMOTE VS. ADASYN UNDER NON-STANDARDIZED PROTOCOL (ACCURACY-ONLY, 10 INDEPENDENT RUNS WITHOUT FIXED SEED).

Run	SMOTE Acc.	ADASYN Acc.	Diff.
1	0.9957	0.9955	+0.0002
2	0.9958	0.9957	+0.0001
3	0.9960	0.9956	+0.0004
4	0.9961	0.9957	+0.0004
5	0.9959	0.9956	+0.0002
6	0.9960	0.9958	+0.0002
7	0.9957	0.9955	+0.0002
8	0.9961	0.9955	+0.0006
9	0.9958	0.9955	+0.0002
10	0.9958	0.9956	+0.0002
Mean	0.9959 ± 0.0002	0.9956 ± 0.0001	+0.0003

V. DISCUSSION

The experimental results presented in Section IV directly address the two research questions formulated in the introduction. Regarding **H1**, the results reported in Tables III–VII indicate that SMOTE, SMOTE+Tomek Links, ADASYN, and class weighting are statistically indistinguishable in both mean performance and variance at imbalance ratios of 1:50 and 1:100 (Wilcoxon $p \geq 0.31$, Levene $p \geq 0.94$). Regarding **H2**, the noise sensitivity analysis (Tables V–VI) shows that all evaluated oversampling strategies degrade comparably under symmetric label noise (≥70% at $\varepsilon = 10\%$), with no measurable robustness advantage for adaptive generation (ADASYN) or boundary cleaning (SMOTE+Tomek) over plain SMOTE. A cross-cutting finding is that the non-standardized accuracy-only protocol consistently produces an accuracy illusion (≈99.6% for all methods) that conceals all fraud-class performance differences, while the standardized PR-AUC protocol yields a reproducible and interpretable conclusion, regardless of whether the outcome is positive or negative.

The absence of differentiation between methods can be attributed primarily to two structural factors. First, the PCA transformation of features V1–V28 produces compact, well-separated class clusters that eliminate the class boundary ambiguity that Tomek Links cleaning is designed to address and that ADASYN's density adaptation is designed to exploit. In such a geometry, all oversampling strategies generate synthetic samples in equivalent regions of feature space, explaining the numerical identity of SMOTE and SMOTE+Tomek and the negligible gap with ADASYN. This observation is consistent with prior analyses noting that method ranking is sensitive to dataset geometry [4], [11]. Second, under label noise, the very small cardinality of the minority class in the experimental subsample (2 fraud instances) renders any oversampling method equally vulnerable to label corruption: any flipped majority label generates a spurious minority centroid around which synthetic samples are generated, flooding the training set with

mislabeled instances regardless of the adaptive design of the oversampling algorithm.

Despite these findings, several limitations constrain the scope of the conclusions. First, all experiments rely on a single base classifier (Random Forest); the observed equivalences may not generalize to gradient boosting or neural network classifiers. Second, the label noise experiments operate on a sub-sample of only 102 instances (100 normal, 2 fraud), which severely limits the statistical power of the noise analysis; a replication on the full dataset with bootstrapping would be necessary for definitive conclusions. Third, the hyperparameter space of the oversampling methods (SMOTE neighbor count k, ADASYN parameter β) was not explored, and default configurations may under-represent the actual capacity of each method. These limitations indicate caution when generalizing our results beyond the tested configuration.

VI. Conclusion and Future Work

This paper presented a controlled, reproducible evaluation of four class imbalance correction techniques — SMOTE, SMOTE + Tomek, ADASYN, and class weighting — on the Credit Card Fraud Detection dataset. Operating under a stratified 5-fold cross-validation protocol with formal statistical testing (Wilcoxon signed-rank, Levene), our experiments yielded three principal findings.

First, SMOTE and SMOTE + Tomek Links produce numerically identical F1-scores (0.8893 ± 0.0213) on a 1:50 imbalanced regime, because the PCA-derived feature space of the dataset eliminates the class boundary ambiguity that Tomek Links cleaning is designed to address. Second, class weighting and SMOTE are statistically indistinguishable in both mean PR-AUC and variance at a 1:100 ratio ($p_{\text{Wilcoxon}} = 1.00$, $p_{\text{Levene}} = 0.94$), challenging the common assumption that cost-sensitive learning is inherently more stable than oversampling under extreme imbalance. Third, all oversampling methods are equally vulnerable to label noise, with degradation rates of 70–75% under symmetric corruption, negating the anticipated robustness advantage of adaptive methods.

A cross-cutting contribution of this work is the empirical demonstration that evaluation protocol standardization — specifically, the adoption of PR-AUC over accuracy and the use of stratified cross-validation — is necessary to produce reliable and reproducible conclusions, even when those conclusions indicate the absence of a significant difference. The non-standardized accuracy-only protocol consistently produces an accuracy illusion that masks all fraud-class performance differences.

Future work should extend this evaluation to non-PCA-transformed fraud datasets with correlated original features, to gradient boosting and neural classifiers, and to optimized hyperparameter configurations for the oversampling methods. Ensemble strategies combining multiple resampling and cost-sensitive approaches also represent a promising direction for improving both mean performance and stability under extreme imbalance.

AI Tools Disclosure

During the preparation of this manuscript, the authors used AI-based language tools to improve readability and structure. All content was critically reviewed and finalised by the authors, who assume full responsibility for the content of the publication.

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